

SUPPLEMENTARY MATERIALS:

Probabilistic State Parameter-Based Liquefaction Triggering Model

Pedro Reyes, Gonzalo Montalva, Vicente San Martín, and Robb Moss

SUPPLEMENT 1. Reduction of cyclic triaxial resistance scatter by CSR_{15} normalization

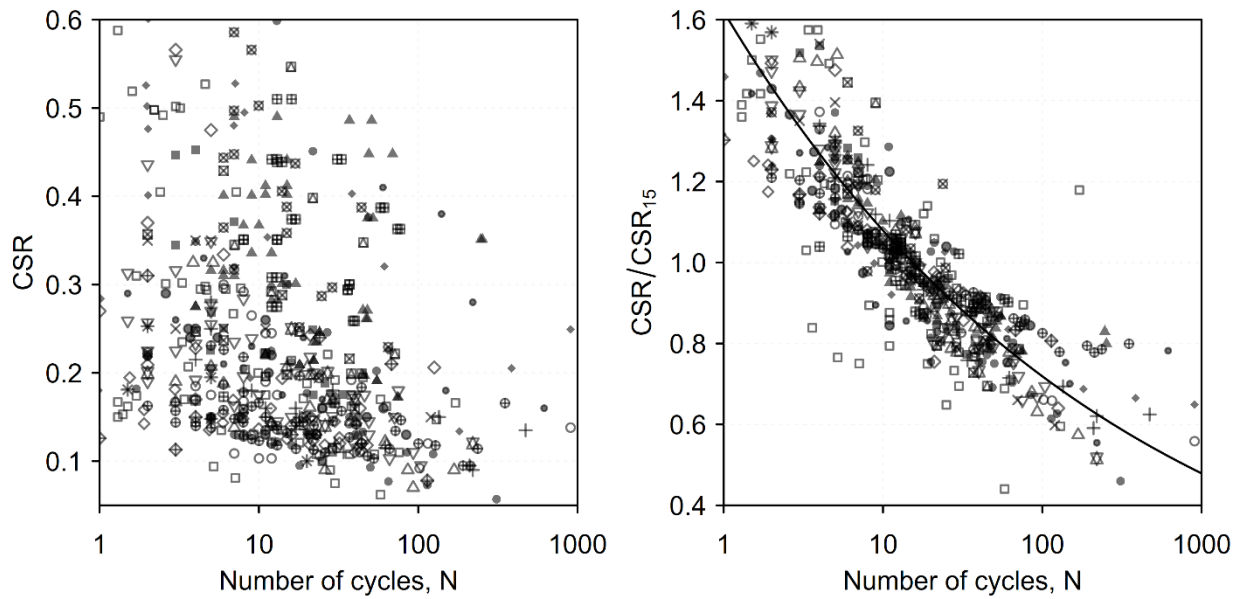


Fig. S1. Cyclic resistance trends from compiled triaxial tests. (a) CSR vs. number of cycles N . (b) normalized response CSR/CSR_{15} vs. N with best-fitting curve using the Jefferies and Been (2015) normalization.

SUPPLEMENT 2. Cyclic resistance uncertainty reduction from material-specific calibration

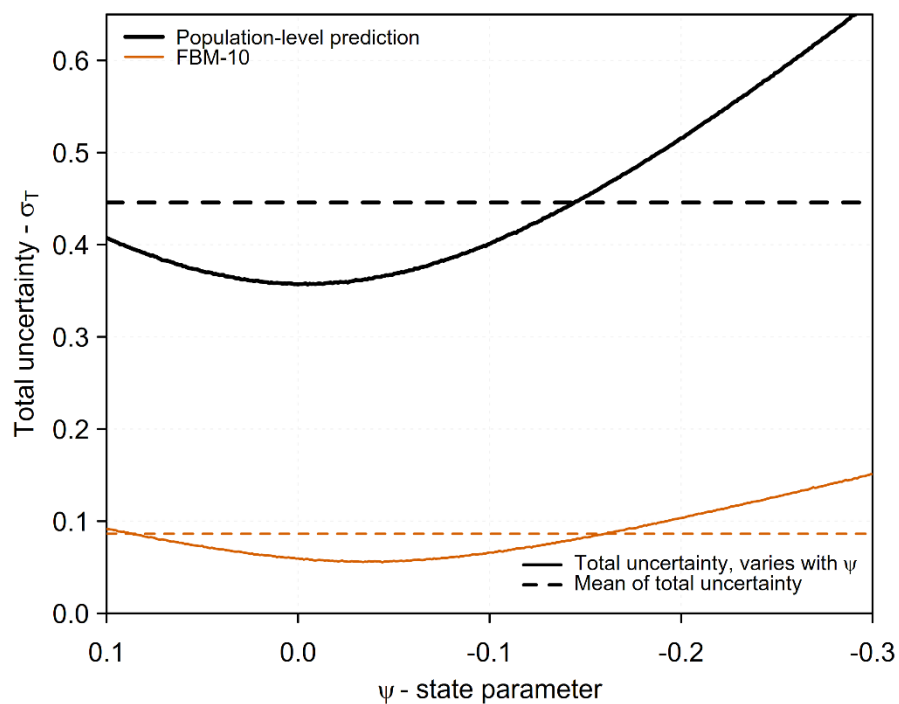


Fig. S2. Total predictive uncertainty, σ_T , as a function of the state parameter, ψ , comparing the population-level prediction and the material-specific calibration (FBM-10).